TECHRISE 3.0

Cybersecurity & DevSecOps Programme · Abia Cohort 3.0 · 2026

PHASE 2 · BLUE TEAM & DEFENCE — FINAL WEEK

WEEK 08

Digital Forensics and Incident Response

Student Training Manual — AI-First Curriculum

PROGRAMME	COHORT	WEEK	PHASE	METHODOLOGY
Cybersecurity & DevSecOps	**Abia Cohort 3.0**	**8 of 12**	**2 — Blue Team & Defence**	**AI-First (Manual → AI)**

Prepared by the TechRise Curriculum Design Team

Lead: Engr. Eduwem Adam · Co-facilitator: Mr. Akorita-Ifeanyichukwu Nehemiah

TABLE OF CONTENTS

- **Week 8: Digital Forensics and Incident Response**
  - Phase: Blue Team and Defence (Weeks 5–8) — Final Week
  - Day 1 — Incident Response Framework (NIST SP 800-61)
    - LEARN · PRACTICE · BUILD · DOCUMENT
  - Day 2 — Chain of Custody and Evidence Preservation
    - LEARN · PRACTICE · BUILD · DOCUMENT
  - Day 3 — Disk Forensics with Autopsy and The Sleuth Kit
    - LEARN · PRACTICE · BUILD · DOCUMENT
  - Day 4 — Memory Forensics with Volatility
    - LEARN · PRACTICE · BUILD · DOCUMENT
  - Day 5 — Review, Assessment & Portfolio (Phase 2 Reflection)
    - LEARN · PRACTICE · BUILD · DOCUMENT
- **Weekly Checklist**
- **Skills Verification**
- **Self-Assessment Quiz**
- **Additional Resources**
- **Looking Ahead — Week 9**

Phase: Blue Team and Defence (Weeks 5–8) — Final Week

Week 8 of 12

WEEK 8: DIGITAL FORENSICS AND INCIDENT RESPONSE

Weekly Overview

This week you learn to investigate security incidents like a forensic investigator. You will master the NIST SP 800-61 incident response framework, preserve evidence with proper chain of custody, capture volatile data from live systems, analyse memory dumps with Volatility, and examine disk images with Autopsy. You will then apply AI to accelerate timeline reconstruction and indicator extraction. By the end of this week, you will have completed the Blue Team phase (Weeks 5–8) and be ready for DevSecOps.

How this week builds on Weeks 5–7

- Week 5 taught you web attacks. Week 6 taught you SIEM detection. Week 7 taught you network detection. Now you investigate what happens after detection — collecting evidence, determining root cause, and preserving artefacts for legal proceedings.
- Your Week 7 PCAP captures become forensic evidence in this week's investigations.

How this week sets up Week 9

- Week 9 is Threat Intelligence and Professional Reporting. The forensic findings you generate this week become the evidence in your professional reports. You will map forensic evidence to MITRE ATT&CK; techniques.

Why this matters in a real security job

- Every security professional needs forensic skills. SOC analysts must collect evidence before rebooting a compromised server. Incident responders must preserve chain of custody for potential legal action. Without forensics, you cannot answer the critical question: 'What happened, and how do we prove it?'

■ Authorisation Reminder

All forensic activities this week target your own lab systems or provided forensic images. Never perform forensics on systems you do not own. Chain of custody is critical for legal admissibility.

DAY 1

Incident Response Framework (NIST SP 800-61)

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Explain the four phases of NIST SP 800-61 incident response

2. Distinguish between events, alerts, and incidents

3. Classify incident severity and determine escalation paths

INCIDENT RESPONSE (IR)

is the systematic process of preparing for, detecting, containing, eradicating, and recovering from security incidents, as defined by NIST SP 800-61 Rev. 2.

NIST SP 800-61 — Four Phases of IR



Events, Alerts, and Incidents — Definitions

Term	Definition	Example	Action
Event	Any observable occurrence in a system	User logged in at 9am	Log it — no action needed
Alert	Event that warrants attention	10 failed logins from same IP	Investigate — may be incident
Incident	Event that threatens CIA triad or violates policy	Successful brute force + root shell	Activate IR process immediately

Incident Severity Classification

Priority	Severity	Example	Response Time	Escalation

P1	Critical	Active ransomware encrypting production	Immediate — all hands	CISO + CEO in 15 minutes
P2	High	Confirmed data exfiltration	1 hour	Manager + Tier 3 in 30 min
P3	Medium	Brute force attack (no breach)	4 hours	Tier 2 investigation
P4	Low	Port scan from external IP	24 hours	Log and monitor

IR Playbook Template

IR Playbook: [Attack Type]

```
■■■
```

08:55	Wazuh: New user 'backdoor' created with UID 0			
08:57	Suricata: Outbound connection to known C2 IP			
09:02	User reports: files are being encrypted on shared drive			

After completing the table: write the escalation communication you would send to your manager for the highest-severity alert. Include: what happened, what you confirmed, what you have already done, and what you need authorisation to do next.

■ ANALYSE THIS

The scenario above represents a complete attack chain. Map each alert to a NIST IR phase AND a Cyber Kill Chain phase. At which alert should you have activated the full IR process? If you had waited until alert 6 (ransomware), what evidence would already be destroyed?

BUILD *(2 Hours)*

BUILD BRIEF

Write two complete IR Playbooks for a fictional Nigerian bank.

Playbook 1 — Ransomware Response:

- Triggering conditions (what alerts activate this playbook)
- Immediate actions (first 15 minutes — what to do before touching anything)
- Containment strategy (isolate without destroying evidence)
- Communication plan (internal: CISO, Legal, IT; external: NDPC 72-hour notification, customers, CBN)
- Recovery sequence (restore order, verification steps)
- Evidence preservation checklist

Playbook 2 — SQL Injection Data Breach:

- Same structure as Playbook 1
- Include: NDPA Article 40 — mandatory breach notification requirements
- Include: CBN data breach reporting requirements for financial institutions

Format:

- Professional operational playbooks. A Tier 1 analyst should be able to follow them during an active incident without needing to think — just execute.

DOCUMENT

Writing Prompt:

Write an Incident Response Lessons Learned Report for the following scenario: A junior developer at QuickPayNG accidentally committed database credentials to a public GitHub repository. The credentials were active for 6 hours before being discovered. The database was accessed by an unknown party during that window. Length: 400–500 words.

- Required sections: Incident Summary, Timeline, Root Cause Analysis, Impact Assessment, What Went Well, What Could Be Improved, Recommended Controls to Prevent Recurrence
- NDPA compliance check: does this constitute a reportable breach? What must QuickPayNG do?

DAY 2

Chain of Custody and Evidence Preservation

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Explain the order of volatility and why it determines collection sequence

2. Create a forensically sound disk image and verify its integrity with SHA256

3. Document chain of custody for collected evidence

CHAIN OF CUSTODY

is the chronological documentation of the collection, transfer, analysis, and disposition of digital evidence — proving that the evidence has not been modified, tampered with, or contaminated between collection and presentation in court.

Order of Volatility — Collect Most Volatile First

```
■■■
```

A forensic image is a bit-for-bit copy of a storage device — including deleted files, unallocated space, and filesystem metadata. It must be:

- Created with a write blocker (prevents any writes to the original)
- Hashed before and after (SHA256 hash must match — proves no modification)
- Documented in chain of custody (who collected, when, from where)

BASH — DISK IMAGING

```
# Step 1: Hash the original disk BEFORE imaging sha256sum /dev/sdb > /evidence/original_hash.txt # Step 2: Create bit-for-bit copy with dd dd if=/dev/sdb of=/evidence/case001_disk.img bs=512 conv=noerror,sync # Step 3: Hash the image file sha256sum /evidence/case001_disk.img >> /evidence/image_hash.txt # Step 4: Verify hashes match # Compare original_hash.txt with image_hash.txt — they MUST be identical # dcfldd: forensic dd with built-in hashing dcfldd if=/dev/sdb of=/evidence/case001.img hash=sha256 hashwindow=1G hashlog=/evidence/hashlog.txt
```

Chain of Custody Documentation

Field	Example	Why It Matters
Case Number	TechRise-2024-001	Unique identifier for all evidence in this case
Evidence ID	E001-DISK	Unique ID for this specific piece of evidence
Description	Western Digital 1TB HDD, Serial WD12345	Identify the exact physical device
Collection Date/Time	2024-01-15 09:47:23 WAT	When evidence was collected
Collected By	Analyst: Akorita Nehemiah, Badge: TR-042	Who touched the evidence
Collection Location	Server Room B, Rack 3, Position 12	Where it was taken from
Condition	Device powered off; no visible physical damage	State of evidence at collection
SHA256 Hash	a3f8...b2c1 (original); a3f8...b2c1 (image)	Proof of integrity
Transfer Record	Transferred to Forensic Lab at 10:15 WAT by...	Track every hand the evidence passes through

WHY THIS MATTERS

In 2016, a major financial institution dismissed a forensics case against an insider threat because the chain of custody was broken — an analyst had copied files without documenting the transfer. The attacker walked free despite digital evidence of the fraud. In Nigeria, the Cybercrimes Act 2015 requires properly authenticated electronic evidence for prosecution. Without chain of custody, your evidence is worthless in court.

PRACTICE *(2 Hours)*

LAB 8.2.1 — Live Response and Evidence Collection

Objective: Perform a structured live response on your Kali VM and document chain of custody.

Prerequisites: Kali Linux running; external USB or dedicated folder for evidence

Step 1 — Create Evidence Directory

BASH

```
# Create timestamped evidence directory CASE="TechRise-$(date +%Y%m%d-%H%M%S)" mkdir -p
~/evidence/$CASE echo "Case started: $(date)" >> ~/evidence/$CASE/chain_of_custody.txt echo "Analyst:
[Your Name]" >> ~/evidence/$CASE/chain_of_custody.txt echo "System: $(hostname)" >>
~/evidence/$CASE/chain_of_custody.txt
```

Step 2 — Document Collection Start

BASH

```
# Record system time (critical for timeline) echo "Collection start: $(date -u)" >>
~/evidence/$CASE/chain_of_custody.txt # Collect volatile data FIRST (memory → network → processes)
ps aux > ~/evidence/$CASE/01_processes.txt ss -tuln > ~/evidence/$CASE/02_network_connections.txt ip
neigh > ~/evidence/$CASE/03_arp_cache.txt ip route > ~/evidence/$CASE/04_routing.txt who >
~/evidence/$CASE/05_logged_in_users.txt last | head -50 > ~/evidence/$CASE/06_login_history.txt cat
/etc/passwd > ~/evidence/$CASE/07_users.txt find / -perm /4000 2>/dev/null >
~/evidence/$CASE/08_suid_files.txt cat /etc/crontab > ~/evidence/$CASE/09_cron.txt # Hash all
collected files sha256sum ~/evidence/$CASE/*.txt > ~/evidence/$CASE/hashes.txt echo "Collection
complete: $(date -u)" >> ~/evidence/$CASE/chain_of_custody.txt
```

Step 3 — Create Test Disk Image

BASH

```
# Create a 100MB test 'disk' to practice imaging dd if=/dev/zero of=/tmp/test_disk.img bs=1M count=100
# Format as ext4 mkfs.ext4 /tmp/test_disk.img # Mount and create test files sudo mkdir
/mnt/test_evidence sudo mount -o loop /tmp/test_disk.img /mnt/test_evidence echo 'Sensitive data' |
sudo tee /mnt/test_evidence/secret.txt echo 'Attack log' | sudo tee /mnt/test_evidence/attack.log #
Delete one file (simulate attacker cleanup) sudo rm /mnt/test_evidence/attack.log sudo umount
/mnt/test_evidence # Hash the 'disk' before imaging sha256sum /tmp/test_disk.img >
~/evidence/$CASE/original_disk_hash.txt # Create forensic image dd if=/tmp/test_disk.img
of=~/evidence/$CASE/forensic_image.img bs=512 # Hash the image (must match original) sha256sum
~/evidence/$CASE/forensic_image.img >> ~/evidence/$CASE/original_disk_hash.txt cat
~/evidence/$CASE/original_disk_hash.txt
```

■ ANALYSE THIS

Both SHA256 hashes should be identical — proving your image is forensically sound. What would it mean if the hashes were different? What would happen to your case in court? How does a write blocker prevent hash mismatch during collection?

BUILD *(2 Hours)*

BUILD BRIEF

Create a Digital Forensics Evidence Kit — all documentation templates needed for a professional investigation.

Template 1 — Chain of Custody Form:

- Case number, Evidence ID, Device description, Collection details, Transfer log, Hash verification section

Template 2 — Live Response Checklist:

- Pre-collection steps, collection order (volatile first), commands to run, output files to save, hash verification

Template 3 — Disk Imaging Log:

- Source device details, write blocker used, imaging tool and version, hash before/after, verification result

Template 4 — Incident Timeline Log:

- Columns: Timestamp (UTC) | Source | Event Type | Description | Evidence Reference | Analyst Note

Format:

- Professional fillable templates. Header on each page: Case Number, Date, Analyst, Page X of Y

These templates should be ready to print and use in a real investigation

DOCUMENT

Writing Prompt:

Write a Digital Evidence Admissibility Guide for Nigerian security professionals.

- Required: Nigeria Cybercrimes Act 2015 — Section 38 on electronic evidence admissibility, Evidence Act 2011 — computer-generated evidence requirements, what makes digital evidence admissible vs inadmissible in Nigerian courts, practical checklist for ensuring evidence will be accepted
- Include a real case where digital evidence was rejected and why

DAY
3

Disk Forensics with Autopsy and The Sleuth Kit

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Navigate Autopsy's interface and create a forensic case

2. Recover deleted files from a disk image

3. Extract browser history, recent files, and user activity artefacts

DISK FORENSICS

is the examination of a forensic disk image to recover evidence — including deleted files, hidden data, file metadata, browser history, and application artefacts — without modifying the original evidence.

The Sleuth Kit / Autopsy Architecture

```
■■■
```

$MFT (Windows)	C:\$MFT	Master File Table — metadata for every file ever created
Prefetch (Windows)	C:\Windows\Prefetch\	Evidence of program execution — even if deleted
Unallocated Space	Between partitions	Deleted files not yet overwritten

TSK Command-Line Reference

BASH — THE SLEUTH KIT COMMANDS

```
# Display partition layout mmls forensic_image.img # List all files (including deleted — marked with *) fls -r -l forensic_image.img # Extract file by inode number (replace 13 with your inode from fls) icat forensic_image.img 13 > recovered_file.txt # Recover ALL deleted files tsk_recover -e forensic_image.img /tmp/recovered_files/ # Search for keywords in image grep -a 'password' forensic_image.img # Extract filesystem statistics fsstat forensic_image.img # List files with timestamps (for timeline) fls -m / -r forensic_image.img > timeline_bodyfile.txt mactime -b timeline_bodyfile.txt > timeline.txt
```

WHY THIS MATTERS

In the 2022 Ronin Bridge crypto hack ($625 million stolen), blockchain forensics and traditional disk forensics combined to identify North Korean Lazarus Group operatives. Forensic investigators recovered deleted SSH keys, command history, and network connection logs from compromised systems. The investigation took weeks — but the forensic trail was there because proper logging and evidence preservation had been configured in advance.

PRACTICE *(2 Hours)*

LAB 8.3.1 — Disk Forensics with TSK and Autopsy

Objective: Recover deleted files from your test disk image and document findings.

Prerequisites: Test disk image from Day 2 lab; TSK installed (sudo apt install sleuthkit)

Step 1 — Hash the Evidence

BASH

```
sha256sum ~/evidence/*/forensic_image.img
```

Record the hash. This proves you are working on an authenticated copy.

Step 2 — List Partitions

BASH

```
mmls ~/evidence/*/forensic_image.img
```

Expected output: shows partition table. Note the starting sector of the main partition.

Step 3 — List Deleted Files

BASH

```
# List all files including deleted (marked with *) fls -r -l ~/evidence/*/forensic_image.img # You
should see: # r/r * 13: attack.log ← the file we deleted! * = deleted
```

Step 4 — Recover the Deleted File

BASH

```
# Extract by inode number (replace 13 with your inode from fls) icat ~/evidence/*/forensic_image.img
13 > ~/evidence/recovered_attack.log # Verify content was recovered cat
~/evidence/recovered_attack.log
```

Expected: 'Attack log' — the content of the deleted file. This is how forensic investigators recover evidence attackers thought they deleted.

Step 5 — Create a Timeline

BASH

```
fls -m / -r ~/evidence/*/forensic_image.img > /tmp/bodyfile.txt mactime -b /tmp/bodyfile.txt >
~/evidence/timeline.txt cat ~/evidence/timeline.txt
```

The timeline shows every file's creation, modification, and access times — critical for reconstructing what happened and when.

Step 6 — Open in Autopsy (GUI)

Launch Autopsy: autopsy & → Open browser → http://localhost:9999. Create new case → Add your forensic_image.img as the data source. Explore: File Analysis tab (look for deleted files), Timeline tab (visual timeline), Keyword Search (search for 'attack').

■ ANALYSE THIS

You recovered the deleted 'attack.log' file. The attacker believed they had covered their tracks by deleting it. What would have permanently destroyed this evidence? (Hint: think about file system overwriting). What should a SOC do to prevent an attacker from destroying evidence before forensics can be performed?

BUILD *(2 Hours)*

BUILD BRIEF

Create a Disk Forensics Investigation Report for the test image from today's lab.

Format: Professional forensic investigation report

Required sections:

- Case Summary: case number, analyst, date, scope
- Evidence Description: disk image details, hash verification result
- Methodology: tools used, analysis steps performed
- Findings:
- - File System Summary: total files, deleted files, filesystem type

- - Deleted File Recovery: what was recovered, content, significance
- - Timeline Analysis: chronological sequence of file system events
- - Key Artefacts: any files of investigative interest
- Conclusions: what the evidence tells you about what happened
- Appendix: hash values, TSK command output, Autopsy screenshots

Quality standard:

- This report should be professional enough to present to a legal team or law enforcement

DOCUMENT

Writing Prompt:

Write a Forensic Analysis Methodology document for your organisation.

- Required: Definition of digital forensics in your organisation's context, when forensics is triggered (what events require forensic investigation), your forensic toolkit (tools, versions, validated against known good), acquisition methodology (live response → imaging → analysis sequence), validation procedures (how you verify evidence integrity), reporting standards
- Include: how you handle evidence from cloud services (AWS, Azure) where you cannot physically seize hardware

DAY 4

Memory Forensics with Volatility

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Explain why memory forensics is essential for detecting fileless malware

2. Run Volatility 3 against a memory dump to extract processes and network connections

3. Identify indicators of malware in memory (injected code, suspicious processes)

MEMORY FORENSICS

is the examination of a captured memory dump (RAM image) to recover volatile artefacts — including running processes, network connections, decrypted data, encryption keys, and malware that exists only in memory (fileless malware).

Why Memory Forensics — What Lives Only in RAM

```
■■■
```

windows.hashdump	vol.py -f mem.dmp windows.hashdump	Extract NTLM password hashes from memory (pass-the-hash)
linux.pslist	vol.py -f mem.dmp linux.pslist	Linux process list
linux.netstat	vol.py -f mem.dmp linux.netstat	Linux network connections
linux.bash	vol.py -f mem.dmp linux.bash	Bash history from memory (even if ~/.bash_history was cleared)

Suspicious Indicators in Memory

Indicator	What It Means	Example
Process with no parent	Orphaned process — may be injected	svchost.exe with PPID=1
Legitimate name, suspicious parent	Process hijacking	cmd.exe spawned by Word.exe
MZ header in non-executable memory	Code injection / shellcode	malfind finds PE header in heap
Network connection to unusual IP	C2 communication	chrome.exe connecting to 185.220.101.5:4444
Process hiding from process list	Rootkit	PID visible in pslist but not pstree
Meterpreter in memory	Active exploitation	rundll32.exe with no disk-based DLL
High entropy sections	Packed/encrypted payload	Unusual entropy in process memory region

WHY THIS MATTERS

The 2017 WannaCry ransomware encrypted 230,000 computers in 150 countries. Memory forensics performed by researchers extracted the WannaCry encryption key from RAM — because RSA decryption keys must be held in memory to function. This enabled development of WannaCry decryption tools (WanaKiwi) that saved files for victims who had not yet rebooted. Memory forensics literally saved millions of dollars of encrypted data.

PRACTICE *(2 Hours)*

LAB 8.4.1 — Memory Forensics with Volatility 3

Objective: Analyse a memory dump and identify malware artefacts.

Prerequisites: Volatility 3 installed (pip3 install volatility3); sample memory dump

Step 1 — Install Volatility 3

BASH

```
pip3 install volatility3 --break-system-packages # Verify installation python3 -m volatility3 -h |
head -5 # Alternative: clone from GitHub git clone
https://github.com/volatilityfoundation/volatility3.git cd volatility3 && pip3 install -r
requirements.txt --break-system-packages
```

Step 2 — Download a Sample Memory Dump

BASH

```
# Download a sample memory dump (380MB) wget
https://github.com/volatilityfoundation/volatility/files/10744946/WinXPSP2x86.vmem.zip unzip
WinXPSP2x86.vmem.zip # Note: This is a large file. Alternatively, use a local capture: # On Linux:
sudo python3 /path/to/LiME/lime.ko 'format=lime path=/tmp/mem.lime'
```

Step 3 — Determine OS Profile

BASH

```
python3 -m volatility3 -f WinXPSP2x86.vmem windows.info
```

Step 4 — Find Hidden Processes (Rootkits)

BASH

```
# List processes python3 -m volatility3 -f WinXPSP2x86.vmem windows.pslist # Process tree (shows
parent-child) python3 -m volatility3 -f WinXPSP2x86.vmem windows.pstree # Compare both — PIDs in
pslist but not pstree = potential rootkit
```

Step 5 — Network Connections

BASH

```
python3 -m volatility3 -f WinXPSP2x86.vmem windows.netstat # Look for: external IPs on unusual ports
(4444 = Meterpreter default)
```

Step 6 — Find Injected Code (Malware)

BASH

```
# Find processes with injected code python3 -m volatility3 -f WinXPSP2x86.vmem windows.malfind #
Output shows: PID, process name, memory region, hex dump of injected code # MZ header (4D 5A) =
Windows PE executable injected into process
```

Step 7 — Extract Command History

BASH

```
# Windows command history from memory python3 -m volatility3 -f WinXPSP2x86.vmem windows.cmdline #
Linux bash history from memory (survives even if file cleared) # python3 -m volatility3 -f linux.mem
linux.bash
```

■ ANALYSE THIS

Malfind found a process with an MZ header (Windows PE executable) injected into its memory space. What attack technique does this represent? Which MITRE ATT&CK; technique ID covers process injection? What is the attacker's goal when injecting code into a legitimate process like explorer.exe?

BUILD *(2 Hours)*

BUILD BRIEF

Create a Memory Forensics Investigation Report for the sample dump.

Required sections:

- Evidence Description: memory dump file, size, SHA256 hash, OS identified
- Process Analysis: complete process list with suspicious processes highlighted (justify why suspicious)
- Network Connections: all active/recent connections with IP reputation (check VirusTotal for external IPs)
- Injected Code Findings: malfind output analysis — for each finding, classify as: malicious/suspicious/benign
- Command Line Evidence: what commands were run? Anything suspicious?
- Indicators of Compromise: extract all IoCs (IPs, process names, injection signatures)
- MITRE ATT&CK; Mapping: map each finding to a MITRE ATT&CK; technique
- Conclusions: overall assessment — was this system compromised? By what?

Format:

- Professional memory forensics report. Evidence-based conclusions only.

DOCUMENT

Writing Prompt:

Write a comparison: Traditional Forensics vs Cloud Forensics.

- Required: What changes when evidence is in AWS/Azure instead of on a physical server, how to collect volatile data from a cloud instance (AWS EC2 memory snapshot, Azure VM capture), chain of custody challenges in multi-tenant cloud environments, legal jurisdiction issues (data in Lagos vs data in US AWS region), tools for cloud forensics (AWS CloudTrail, Azure Monitor, GCP Cloud Audit Logs)
- This is increasingly relevant as Nigerian enterprises move to cloud

DAY
5

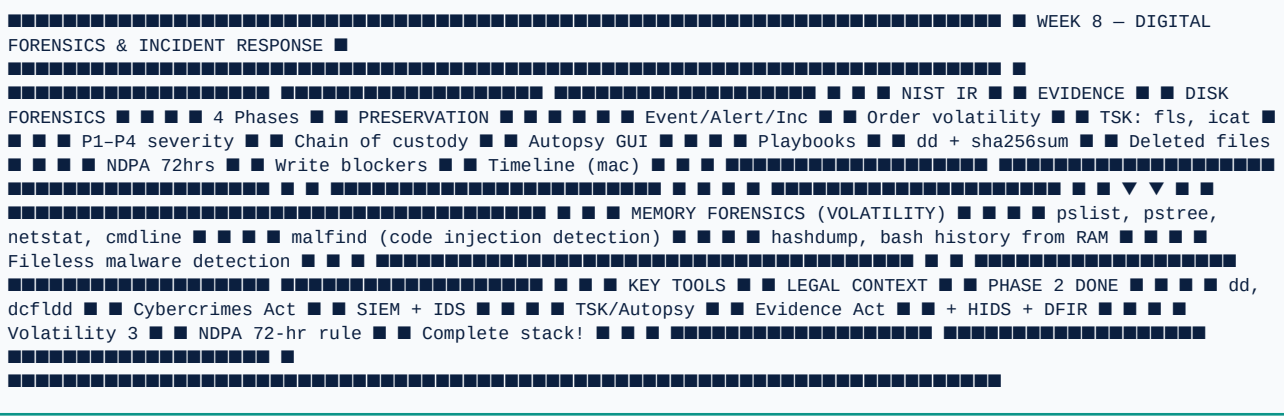
Review, Assessment & Portfolio (Phase 2 Reflection)

LEARN *(1 Hour) — Week and Phase Review*

LEARNING OBJECTIVES

1. Demonstrate all Week 8 forensics skills without referencing notes
2. Complete the Phase 2 portfolio entry
3. Reflect on the complete Blue Team phase (Weeks 5–8)

Week 8 Mind Map — Digital Forensics & IR



PRACTICE *(2 Hours) — Skills Assessment Labs*

LAB 8.5.1 — Rapid Response Simulation

Objective: Respond to a simulated incident from detection to evidence collection — 45 minutes.

Prerequisites: All tools from Weeks 6–8; no notes

Scenario: Your Splunk dashboard fires a P2 alert — 'Successful SSH login after brute force from 192.168.1.100 followed by new user creation.' You are the on-call analyst.

1. Classify the incident (severity, type) and write your escalation note
2. Perform live response on the 'compromised' system (your Kali VM) — capture volatile data in correct order of volatility
3. Check Splunk for full attack timeline — when did brute force start? When was new user created?
4. Check Suricata/Snort alerts — was there any network traffic to external IPs from this system?
5. Create a forensic disk image of /tmp/test_disk.img and verify hash
6. Run Volatility on any available memory dump — list processes and network connections
7. Write a 1-page Incident Summary covering all findings

LAB 8.5.2 — Forensics Accuracy Test

Objective: Verify your forensics skills by answering questions from a pre-prepared scenario.

Prerequisites: TSK/Autopsy; Volatility; test disk image from Day 2

Answer these questions from your forensic evidence only — no external references:

1. What file was deleted from the test disk image? What was its content?
2. What is the SHA256 hash of your forensic disk image?
3. At what exact timestamp was the deleted file created? (Use mactime output)
4. From your Volatility output — name 3 processes running and their PIDs
5. Write the chain of custody entry for the disk image you created in Day 2

BUILD *(2 Hours) — Phase 2 Portfolio Entry*

- 1. NIST IR Framework — your completed IR playbook for ransomware response
- 2. Chain of Custody — completed chain of custody form from Day 2 live response
- 3. Live Response Output — commands run, files collected, hashes verified (from Day 2)
- 4. Disk Forensics — TSK commands run, deleted file recovered, timeline generated
- 5. Memory Forensics — Volatility plugins run, suspicious processes identified, IoCs extracted
- 6. AI Integration Log — how you used AI during this week, what it found, what you corrected
- 7. Phase 2 Self-Reflection — 300-word reflection on the complete Blue Team phase (Weeks 5–8)

■ PHASE 2 COMPLETE — BLUE TEAM & DEFENCE

By completing Week 8, you have finished the full Blue Team phase. You now have: Complete detection stack (SIEM + IDS + HIDS) · Web attack knowledge and detection · Network forensics capability · Incident response skills · Evidence preservation and analysis. Weeks 9–12 cover Phase 3: DevSecOps, Cryptography, Cloud Security, and Capstone.

DOCUMENT *(1 Hour) — Week Summary & Phase Reflection*

DOCUMENT

Writing Prompt:

Write your Phase 2 Completion Reflection (Weeks 5–8). Length: 400–500 words.

- What is your strongest Blue Team skill after this phase? Demonstrate with a specific example.

- What was the hardest concept across the 4 weeks (OWASP, SIEM, IDS, or DFIR) and how did you overcome it?
- How has your understanding of attack techniques (Phase 1, Weeks 1–4) made you a better defender in Phase 2?
- If you were hired as a Tier 1 SOC analyst tomorrow, what would you do on your first day using the skills from Weeks 5–8?
- What defensive tool or technique from Weeks 5–8 do you want to develop most deeply in your career?

Weekly Checklist (Student-Facing)

- ■ I can explain all 4 phases of NIST SP 800-61 from memory and give one example activity per phase
- ■ I can classify any security event as Event / Alert / Incident and assign P1–P4 severity
- ■ I have written a complete IR playbook for at least one attack type
- ■ I understand the order of volatility and can explain why RAM must be collected before disk
- ■ I have created a forensic disk image with dd and verified it with SHA256
- ■ I have completed a chain of custody form for evidence I collected
- ■ I have recovered a deleted file from a disk image using TSK (icat/fls)
- ■ I have run Volatility against a memory dump and identified running processes
- ■ I have used Volatility's malfind to identify suspicious code injection
- ■ My Week 8 Portfolio Entry and Phase 2 Reflection are complete

Skills Verification

Skill	How to Demonstrate It	Pass Standard
NIST IR phases	Name all 4 phases and one activity per phase without notes	All 4 correct with specific activities
Incident classification	Given 5 alerts, classify each as Event/Alert/Incident with severity	At least 4 out of 5 correctly classified
Evidence collection	Run live response script and show timestamped output files	All volatile data captured in correct order
Disk imaging	Create forensic image and show matching SHA256 hashes	Hash match confirmed; command documented
Deleted file recovery	Recover deleted file from test image using icat	File content recovered; inode documented
Memory forensics	Run pslist and netstat on a memory dump	Processes and connections listed correctly
Malfind analysis	Show malfind output and explain what MZ header indicates	Correct explanation of code injection
Chain of custody	Complete chain of custody form for evidence collected today	All fields filled; hash recorded

Self-Assessment Quiz

Instructions

- No notes — answer from memory.
- Write on separate sheet.
- Key in Instructor Notes.

Conceptual Questions

1. Explain the 'order of volatility.' If you arrive at a compromised server that is still powered on — list the first 5 things you collect and why that specific order.
2. What is the difference between a forensic image and a simple file copy? Why does it matter for deleted file recovery?
3. What is 'fileless malware'? Why can disk forensics alone not detect it? What specific Volatility plugin would you use to find it?

Tool Knowledge

4. Write the exact command to: (a) create a forensic image of /dev/sdb and save it to /evidence/case.img, (b) generate SHA256 hash of the image, (c) list deleted files in the image.
5. You run vol.py windows.malfind and get output showing a process with an MZ header at address 0x400000. Explain: what this means, which attack technique it represents, and what you do next.

Scenario Questions

6. At 2:17 AM your Splunk alert fires: a process called 'svchost.exe' on a domain controller has an active outbound connection to 185.220.101.5 on port 4444. Your Volatility output shows this svchost.exe has no parent process (PPID=0) and malfind shows an MZ header in its memory. Write your incident response — minute by minute for the first 30 minutes.

Legal and Compliance

7. A Nigerian fintech discovers evidence of insider fraud on an employee's workstation. To use this evidence in Nigerian court: what must the forensic analyst have done to ensure the evidence is admissible? Reference the specific Nigerian law. What would make it inadmissible?
8. Your forensic investigation reveals that a customer database was accessed and exfiltrated 6 days ago. Under Nigeria's Data Protection Act 2023: what must the organisation do? What is the deadline? What is the maximum penalty for failing to notify?

Additional Resources

Incident Response

- NIST SP 800-61 Rev. 2 — Computer Security Incident Handling Guide: https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf
- SANS Incident Response Process (Free): https://www.sans.org/score/checklists/
- TryHackMe — Incident Response Room: https://tryhackme.com/room/incidentresponse

Disk Forensics

- The Sleuth Kit Documentation: https://www.sleuthkit.org/sleuthkit/
- Autopsy Digital Forensics Platform: https://www.autopsy.com
- TryHackMe — Autopsy Room: https://tryhackme.com/room/autopsy2ze0
- SANS DFIR Posters (Free Reference): https://www.sans.org/posters/dfir/

Memory Forensics

- Volatility 3 GitHub: https://github.com/volatilityfoundation/volatility3
- Volatility 3 Documentation: https://volatility3.readthedocs.io
- TryHackMe — Volatility Room: https://tryhackme.com/room/volatility
- The Art of Memory Forensics (Book): Ligh et al., 2014

Nigerian Legal Context

- Nigeria Cybercrimes (Prohibition, Prevention, Etc.) Act 2015: https://www.nitda.gov.ng
- Nigeria Evidence Act 2011: https://www.nigeria-law.org
- NDPA 2023 — Nigeria Data Protection Act: https://ndpc.gov.ng
- CBN Data Breach Reporting Guidelines: https://www.cbn.gov.ng

Looking Ahead — Week 9

Preview — What's Coming in Week 9

- Phase 3 begins — DevSecOps, Threat Intelligence, and Professional Reporting
- Threat Intelligence — MITRE ATT&CK;, IoC feeds, and threat actor profiling
- Professional Report Writing — how to present your forensic findings to executives and legal teams
- Your Week 8 forensic findings become professional threat intelligence reports
- MITRE ATT&CK; Navigator — map your entire investigation to the framework

What to Review Before Week 9

- Your complete Week 8 forensic findings — they become Week 9 report content
- MITRE ATT&CK; framework — review the techniques you identified during Weeks 4–8
- Your Splunk detection rules from Week 6 — you will map them to ATT&CK; techniques

What to Set Up Before Week 9

- Create a free MITRE ATT&CK; Navigator account: https://mitre-attack.github.io/attack-navigator/
- Review your Weeks 4–8 notes — Week 9 requires recall of all phases
- Organise your portfolio entries from Weeks 1–8 — Week 9 begins your final portfolio preparation

■ PHASE 2 COMPLETE — BLUE TEAM & DEFENCE

By completing Week 8, you have finished the full Blue Team phase. You now have: Complete detection stack (SIEM + IDS + HIDS) · Web attack knowledge and detection · Network forensics capability · Incident response skills · Evidence preservation and analysis. Weeks 9–12 cover Phase 3: DevSecOps, Cryptography, Cloud Security, and Capstone.